

Qi Kang (Kevin)

[Email](#) | [Google Scholar](#) | [Linkedin](#) | [Website](#)

EDUCATION

NORTHEASTERN UNIVERSITY (BOSTON)

Ph.D. Candidate, Data and Systems

Boston, United States

Sep. 2024 – Expected 2028

- Research focus: Autonomous Driving; Reinforcement learning; Physical AI Safety

NATIONAL UNIVERSITY OF SINGAPORE

Master of Engineering (Thesis), Mechanical Engineering

Singapore

Jul. 2021 – Nov. 2023

- Research focus: Autonomous Truck control, Robust Reinforcement Learning, Multi-robot System

HUAZHONG UNIVERSITY OF SCIENCE AND TECHNOLOGY

Bachelor of Engineering, Mechanical Design Manufacture and Automation

Wuhan, China

Sep. 2017 – Jun. 2021

WORK EXPERIENCE

Continental-NTU Corporation Lab

Research Engineer

Singapore

Aug. 2023 – Aug. 2024

- Developed a multi-agent RL framework for large-scale autonomous fleet rebalancing, coordinating vehicles cooperatively to maximize the fleet's level of service.
- Modelled and predicted human drivers' vacant-travel behavior from operational data with multivariate XGBoost, aligning dispatch decisions with learned driver preferences.

Continental Automotive (Now AUMOVIO)

AI Function Engineer Intern

Singapore

May. 2022 – Apr. 2023

- Developed a PPO-based RL tracking controller for an autonomous tractor-trailer to follow reference paths.
- Built a PyBullet training environment with a custom 3D vehicle model and B-spline randomized tracks; the trained policy followed paths without off-tracking, even at sharp turns (first-author IEEE ITSC 2024 paper).

Agency for Science, Technology and Research (A*STAR)

Research Student Intern

Singapore

Nov. 2022 – Jun. 2023

- Proposed R-FAC (joint first author, IEEE T-RO), a robust multi-agent RL method that keeps a multi-robot search team performing well even when individual robots break down mid-task.
- Developed EM-Patroller (IEEE RA-L), a multi-robot patrolling method with per-iteration improvement guarantees.

SELECTED PUBLICATION (* corresponding author; † joint first author)

1 H. Lin, Y. Liu, **Q. Kang***, H. Huang, Y. Liu, H.N. Koutsopoulos, J. Zhao, "[VeriFuse: Bounded Vision-Language Arbitration and Reason-Guided Refinement for Cooperative 3D Perception](#)" submitted to ICRA 2027, arXiv: 2609.21323

2 **Q. Kang***, H.N. Koutsopoulos, H. Guo, J. Zhao, "[Hi-CoLA: High Confidence Lower Bound Approximation Based Reinforcement Learning for Flex-Route Transit Operation Control](#)," in ICRA 2026 [Workshop](#), Vienna, Austria.

3 H. Guo*†, **Q. Kang†**, W. -Y. Yau, C.-M. Chew and D. Rus, "[R-FAC: Resilient Value Function Factorization for Multi-Robot Efficient Search with Individual Failure Probabilities](#)," in *IEEE Transactions on Robotics*, vol. 41, pp. 3385-3401.

4 **Q. Kang***, A. Hartmannsgruber, S.-H. Tan, X. Zhang and C. -M. Chew, "[Deep Reinforcement Learning Based Tractor-Trailer Tracking Control](#)," in *2024 IEEE 27th International Conference on Intelligent Transportation Systems (ITSC)*

PROJECT EXPERIENCE

Drive2Survive: A Safety Benchmark for VLA-based E2E Autonomous Driving Models

Project Lead — manuscript and leaderboard in preparation

MIT Transit Lab, United States

Jul. 2026 – Present

- Design a safety benchmark that rewards VLA driving planners for defusing pre-crash risk rather than avoiding collisions by chance, built on high-risk scenarios selected from nuPlan human driving logs via crash-precursors.
- Evaluated SOTA VLA planners on NAVSIM v2 with crash-rate and precursor metrics, releasing metric code and a public leaderboard to encourage community engagement.

Safe Reinforcement Learning with High-Confidence Performance Guarantees

Graduate Research Assistant

MIT Transit Lab, United States

Sep. 2024 – Jan. 2026

- Developed a confidence-integrated RL training framework giving formal performance guarantees and monotonic policy improvement for driving and routing agents.
- Applied the framework to dynamic vehicle routing control, improved worst-case policy performance.

Full-Stack Autonomous Driving System — Formula Student Autonomous Racing (HUST)

Leader & Chief Designer in a team of over 60 students

Wuhan, China

Jan. 2019 – Jan. 2020

- Designed the architecture of a complete autonomous driving stack: LiDAR-based point cloud processing for lane/track detection, heuristic trajectory planning, and PID-based motion stabilization.
- Won a National Award at Formula Student Autonomous China (FSAC) with a fully autonomous race car.

SKILLS

Reinforcement Learning, VLA based Trajectory Control, Closed-loop Simulation, VLM, Vehicle Motion Control